Personality Prediction from Moments to Lifetimes

Emorie D. Beck[1]

Adam T. Nissen[1]

Eileen K. Graham[2,3]

Daniel K. Mroczek[2,3]

[1]University of California, Davis, Psychology Department

[2]Northwestern University Feinberg School of Medicine, Department of Medical Social Sciences

[3]Northwestern University, Department of Psychology

Authors' Note. Correspondence should be addressed to Emorie D. Beck, Psychology Department, University of California, Davis, One Shields Ave., Davis, CA 95616; edbeck@ucdavis.edu. Graham and Mroczek are co-senior authors. The preparation of this article was supported by National Institutes of Aging grants RF1-AG088206 (PI: Beck), RF1-AG094591 (MPI: Graham/Luo), R01-AG082954 (PI: Graham), RF1-AG067622 (PI: Mroczek), and U24-AG081810 (MPI: Scott/Neupert) as well as a National Science Foundation Grant BCS-2336406 (co-PI: Beck).

Abstract

Personality is robustly linked to long-term outcomes, yet predicting momentary behavior (i.e., who does what, when, and where) remains challenging. We argue this gap reflects a mismatch between the levels at which personality and outcomes are studied. Both can be organized within a nested, emergent hierarchy spanning macro-, meso-, and micro-levels of analysis. At the macro-level, broad traits (e.g., the Big Five) reliably predict distal outcomes, including cognitive aging and mortality, but offer limited insight into underlying processes. At the meso-level, personality states predict shorter-term outcomes, though individual heterogeneity is often overlooked and findings examined in isolation. At the micro-level, personality processes unfold over short timescales as complex multivariate systems, enabling personalized intervention but often limiting generalizability. To reconcile these trade-offs, we propose a bottom-up integrative framework in which summaries of micro-level personality dynamics are linked to meso- and macro-level outcomes. Emerging evidence suggests such summaries provide a promising bridge across levels, supporting integrative and explainable prediction.

Personality Prediction from Moments to Lifetimes

Personality is a strong predictor of major life outcomes, including longevity, health, wealth, relationship quality, and occupational success (e.g., Ozer & Benet-Martínez, 2006). Yet links between personality and specific behaviors remain elusive (e.g., Beck & Jackson, 2022c). This challenge, raised decades ago by Jerry Wiggins (Wiggins, 1973), stems from assumptions that broad traits should predict narrow behaviors across time and from debates over whether narrow predictors are required for narrow outcomes. We argue that resolving this issue requires moving beyond traits to include states and processes, and that advances in personality dynamics offer a framework for bridging levels of prediction.

We propose that accurate personality prediction requires jointly considering macro-level dispositions, meso-level states, and micro-level processes[i]. This integrated approach can improve prediction of short- and long-term outcomes and support scalable interventions and policy development (Bleidorn et al., 2019). As defined in Table 1, constructs and outcomes span multiple levels of aggregation and timescales, making some better suited for particular outcomes. These constructs are nested (momentary processes underlie states that underlie traits), have fuzzy boundaries [ii] (affect over seconds vs. mood over hours or days), and show bidirectional associations (see Figure 1). This pattern aligns with an emergent hierarchy in which causal forces are distributed across levels, making each essential for understanding personality prediction (Jansma & Hoel, 2025).

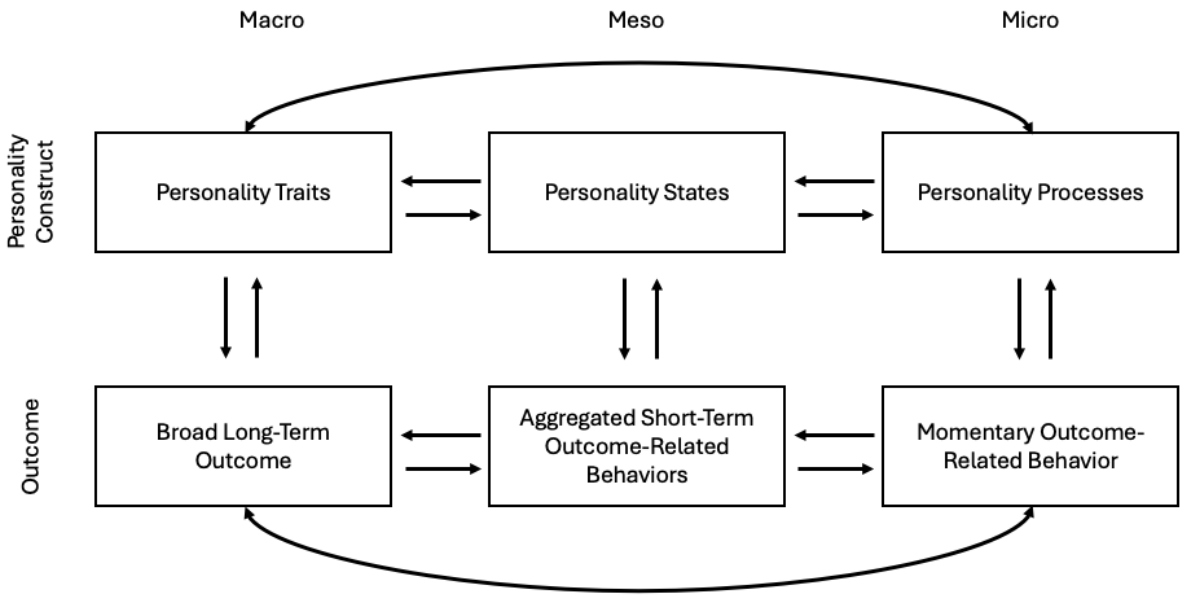
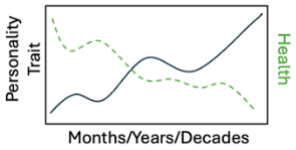
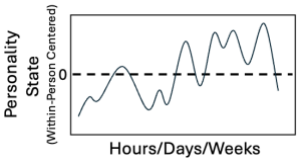
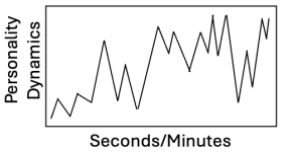


Figure 1. Associations between personality and outcomes across macro, meso, and micro levels

Table 1
Overview and definition of core constructs across timescales and levels of aggregation

Level of Aggreg-ation	(Rough) Time-scale	Graphical Timescale	Personality Constructs	Outcomes
Long-term (Macro)	Months, years, decades		Broad differences in stable, aggregate psychosocial differences (e.g., neuroticism)	Broad outcomes, events, and stable conditions (e.g., mortality, chronic conditions, physical fitness)
Short-term (Meso)	Hours, days, weeks, (months, years)		Behaviors and experiences that have accumulated through some timespan (e.g., negative emotion in the last hour)	Behaviors and experiences that have accumulated though some timespan (e.g., physical activity across a day; health care utilization)
Moments (Micro)	Seconds, minutes, (hours, days)		Momentary behaviors and experiences (e.g., current negative affect or fluctuations of negative affect) and their dynamics	Momentary behaviors and experiences (e.g., physical activity in the moment via heart rate; medical support seeking in a given moment)

The paper proceeds as follows: We review theory and evidence at each level, highlighting personality prediction and health as running examples, and conclude with a generative research paradigm aimed at bridging levels.

Macro Level: Personality Traits and Long-Term Outcomes

Personality traits, or characteristic patterns of thinking, feeling, and behaving, are thought to influence behaviors across timescales that accumulate into consequential life outcomes (Ozer & Benet-Martínez, 2006). Their ability to predict long-term outcomes (e.g., mortality, cognitive aging) likely reflects their influence on meso-level behaviors (e.g., health care usage) and micro-level processes (e.g., stress).

Accordingly, extensive research has examined how traits relate to long-term health and mortality to identify who is at risk for poor health and shorter lifespans (Beck & Jackson, 2022a). Findings consistently show that high conscientiousness and low neuroticism protect against chronic conditions, mortality, and cognitive decline, suggesting personality can serve as an early risk marker. For example, conscientiousness and neuroticism robustly predict dementia diagnoses across eight international samples (Beck et al., 2024), and *increases* in neuroticism also predict dementia (Yoneda et al., 2020).

However, several challenges remain due to complex cross-level connections. First, personality–health associations are not always monotonic across the lifespan; for instance, conscientiousness's positive effects diminish with age, likely due to increasing biological burden (Luo et al., 2023). Second, personality appears more useful for understanding responses to biological burden than predicting burden itself. Studies comparing macro-level traits with micro-level neuropathology rarely find direct links (Beck et al., 2024), yet traits predict cognitive

resilience (performance despite pathology) with higher conscientiousness and lower neuroticism linked to greater resilience (Graham et al., 2021).

Finally, trait–outcome associations may reflect reverse causality (e.g., poor health increasing negative emotions). Mortality offers a clearer directional test; decades of research show that higher neuroticism predicts lower survival, whereas conscientiousness, and to a lesser degree openness and extraversion, protects against early mortality. These effects remain after adjusting for demographics and replicate worldwide (Graham et al., 2017; Jokela et al., 2013; Willroth et al., 2025). Combined with inconsistent evidence that long-term outcomes shape personality (e.g., Bleidorn et al., 2018), this work suggests personality is more influential than the reverse.

Summary: Promises and Challenges of the Macro-Level

Macro-level research demonstrates robust links between traits and outcomes. Because self-report trait measures are efficient and economical, this work offers practical value for identifying broad health risks. Yet macro-level findings mirror *the therapist's dilemma*: Group-level associations identify who is at risk but not why these links emerge, often requiring meso- and micro-level accounts crucial for intervention (Levine et al., 1992).

Meso-Level: Personality States and Short-Term Outcomes

Relative to the macro level, the meso level focuses on aggregated behaviors and experiences unfolding over shorter timescales (e.g., weeks, months) that are theoretically and empirically critical. Theoretically, accumulated behavior is assumed to (1) link traits to long-term outcomes (Ozer & Benet-Martínez, 2006) and (2) drive personality change (Wrzus & Roberts, 2017). Empirically, repeated meso-level measures address the ecological fallacy by

separating *between-person differences* from *within-person associations* over time, which may differ in direction, and by allowing investigation of individual differences in these within-person links. For example, although neuroticism predicts poorer long-term health, within-person increases may be associated with either protective behaviors (e.g., preventive care) or harmful behaviors (e.g., substance use), depending on the individual.

Despite this relevance, meso-level personality prediction remains limited. First, much research straddles macro and meso levels by asking who engages in meso-level behaviors rather than when. For instance, traits predict health care utilization (HCU): Higher conscientiousness, extraversion, and openness, and lower neuroticism, are linked to preventive HCU, whereas lower conscientiousness and agreeableness and higher neuroticism predict hospitalization risk (Atherton et al., 2024; Weston et al., 2024; Willroth et al., 2023). Such findings clarify who uses care but not the conditions under which they do so. Our work with middle-aged and older adults illustrates this gap: Lower trait conscientiousness predicts poorer average cognitive performance, yet moments of higher conscientiousness predict better performance (Ma et al., under review), indicating that macro-level findings can misguide meso-level intervention.

Second, particularly in health research, meso-level studies often rely on long timescales (years or decades) that do not match the processes of interest (Hopwood et al., 2022). For example, better health is reported in years when individuals are less neurotic and more extraverted and conscientious than usual (Luo et al., 2022), yet shorter-term fluctuations (e.g., daily), which may distinguish individuals with similar trait levels, are rarely observed. The rise of intensive longitudinal data (e.g., ecological momentary assessment) enables prediction in daily life; for instance, moments of higher conscientiousness and lower extraversion predict better

momentary cognitive performance (Ma et al., under review), highlighting actionable timescales for intervention.

Third, within-person studies reveal substantial individual differences in *when* and *for whom* personality states predict health behaviors, yet these differences are rarely centered. Because predictors of between-person variation are often untested, it remains unclear whether individuals showing certain within-person patterns simply differ due to psychosocial or demographic factors. Although recent work has examined between-person moderators of within-person associations (i.e., cross-level interactions), this literature has focused primarily on demographic rather than personality moderators, pointing to an important direction for future research.

Summary: Promises and Challenges of the Meso-Level

The meso-level captures short-term aggregates of behaviors and experiences and enables modeling of joint between- and within-person associations that can translate into practical intervention targets. However, complex analytical models (e.g., multilevel, cross-lagged panel) hinder examination of multiple within-person processes and interactions. The next section highlights how micro-level processes studied within a systems framework may help bridge micro-, meso-, and macro-levels.

Micro-Level: Heterogenous Personality Processes and Momentary Behavior

While personality traits show robust links to meso- and macro-level outcomes, accurately predicting momentary, micro-level outcomes remains elusive. Effect sizes for short-term behaviors and long-term events are relatively small, partly because outcomes arise from a complex web of contributing factors (Götz et al., 2022), much of which remains untested due to

methodological constraints. Advances in methods (e.g., EMA, ML/AI) and theory (e.g., dynamic systems) are enabling new progress in micro-level prediction, particularly in forecasting *whether and when* individuals enact specific behaviors (Beck & Jackson, 2022c).

Dynamics approaches conceptualize personality as a multivariate, person-specific system of affective, cognitive, and goal processes unfolding over time and context (Beck & Jackson, 2020b). This systems perspective offers several advantages: it (1) treats traits as emergent properties of heterogeneous processes (e.g., Baumert et al., 2017), (2) targets the whole person, (3) models complex interactions among processes (Beck & Jackson, 2020a; Lee & Beck, 2025), and (4) incorporates person–environment interactions (Beck & Jackson, 2020b). Dynamics approaches aim to understand personality bottom-up—capturing individual differences in heterogeneous processes—and hold promise for creating cross-sectional measures that better map onto momentary dynamics, an approach termed bottom-up integration (Nissen & Beck, 2025b).

Research on personality dynamics and intensive longitudinal research are too broad to review fully here. Below, we highlight the promise of systems approaches for predicting micro-level outcomes and show how system properties can enhance prediction at meso- and macro-levels.

Person-Specific Ensembles of Person- and Situation Factors Predict Behavior

A long-standing question in behavior prediction concerns the relative importance of person versus situation factors (Mischel, 1968). Although both show similar effect sizes across studies, evidence for person × situation interactions (i.e. individuals with certain traits responding differently to context) remains sparse (Kuper et al., 2024). Rather than asking which factor "wins" on average, the crucial question is when and for whom each matters (Beck &

Jackson, 2022c), requiring a multivariate approach accommodating heterogeneity across individuals.

Beck and Jackson (2022c) tested this using person-specific machine learning to identify when person, situation, or both factors best predicted behavior. Figure 2 shows that even the same behavior can stem from different antecedents (equifinality) across individuals: some are best predicted by (A) both person and situation, others by (B) situation, (C) person, or (D) single-feature associations. Figure 2E demonstrates that this heterogeneity occurs both across people and within individuals across behaviors.

In testing whether (1) seven future (t+1) behaviors and experiences were predictable from person and situation factors, (2) individuals differed in prediction strength, and (3) the most influential features varied across people, they found prediction accuracy far above chance and typically observed trait–outcome associations. Individuals also differed substantially in both the relative importance of person and situation features and the specific features most predictive of their behaviors.

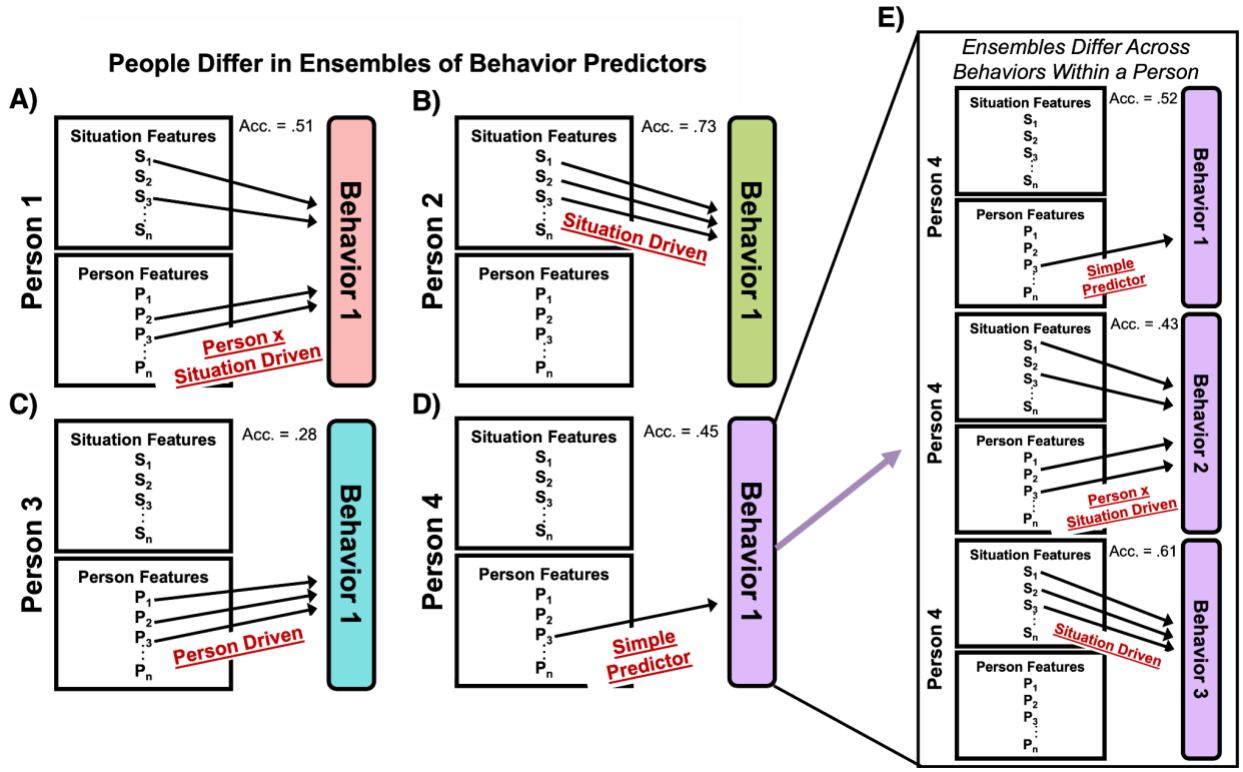


Figure 2. Examples of different patterns of person (P_1 to P_n) and situation (S_1 to S_n) factors predicting the same behavior across people (panels A-D) and different behaviors for one person (panel E). Each arrow represents a predictor-outcome association that was retained in a machine learning model. Acc. = out of sample prediction accuracy (range 0-1).

Thus, modest historical success in micro-level behavior prediction likely stems from oversimplified, mechanistic models that overlook behavior's inherently multi- and equifinal nature. Yet, key gaps remain, most critically how heterogeneous multivariate feature ensembles from machine learning relate to aggregated behavioral expression (meso-level states) and long-term characteristics (macro-level traits). Combined with evidence that personality can be operationalized as heterogeneous, person-specific multivariate time-series networks (Beck & Jackson, 2020a), these findings imply that personality is weakly emergent and non-ergodic, meaning micro-level processes cannot be inferred from meso- or macro-level patterns. Consequently, linking macro-level traits to outcomes through (linear) meso-level accumulation is likely too simplistic to explain why such relationships arise. Thus, how can we more effectively

connect micro-, meso-, and macro-level personality–outcome associations to deepen our understanding of each?

Integrating From the Micro to the Macro (And Vice Versa): Personality Systems

Given the near-infinite combinations of potential antecedents, a common critique is that a person-specific systems approach lacks utility because it does not generalize across people. Yet, we argue that dynamic systems and network-theoretic approaches offer an integrative framework for summarizing complex psychological systems in ways that permit comparisons across individuals despite qualitative and quantitative heterogeneity (see Figure 3). Put differently, our approach asks whether summaries of heterogeneous models improve long-term outcome prediction while still preserving insight into micro-level processes and dynamics?

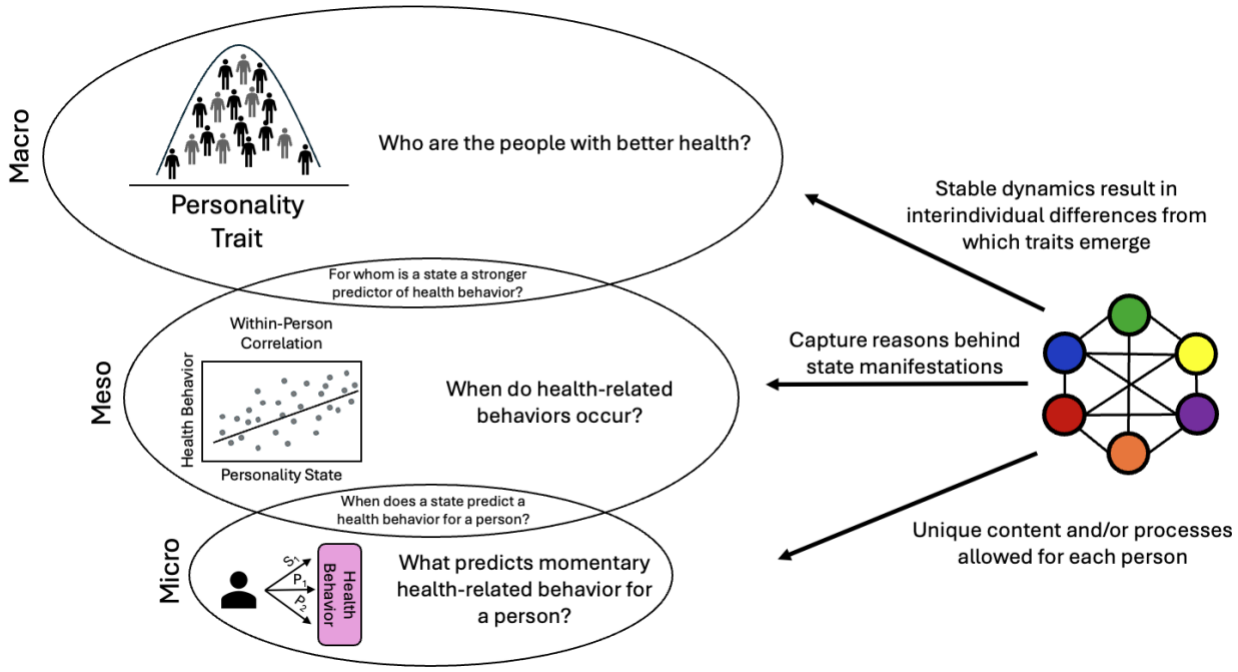


Figure 3. *N*=1 personality networks as a proxy for personality systems for understanding prediction across levels.

Personality Systems are Associated with Distal Outcomes and Trait Change

In a series of two studies, Nissen and colleagues operationalized micro-level person-specific personality systems as *N*=1 time series networks of Big Five personality states and investigated how properties of these networks were associated with macro-level, long-term personality trait change (Nissen & Beck, 2025a) and long-term outcomes (Nissen et al., under review). From a dynamic systems perspective, several features of systems make them more likely to demonstrate change or experience outcomes. For example, vulnerability through interconnectivity suggests that denser connections in a network are more vulnerable to substantive long-term changes (Beck & Jackson, 2022b).

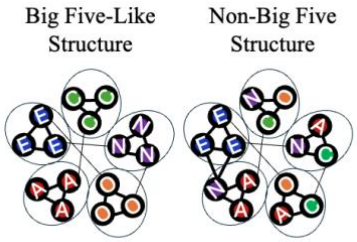
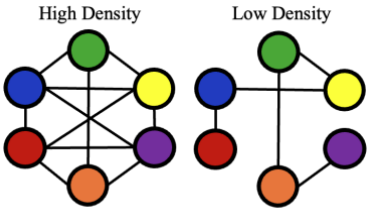
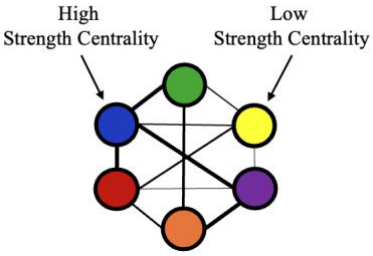
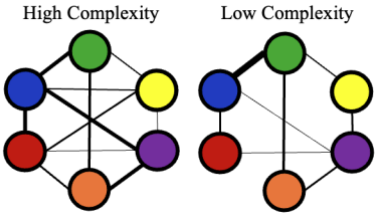
Properties included network-theoretic parameters capturing organization (e.g., normalized mutual information), topology (e.g., density), and complexity (e.g., entropy; see Table 2). They examined whether these network-level properties predicted Big Five trait change over two years (Nissen & Beck, 2025a) and 13 distal life outcomes (Nissen et al., under review). Unlike purely macro-level approaches, this design still permits inferences about single-case dynamics, offering insight into mechanisms underlying these associations. Network parameters were largely unrelated to changes in self-reported traits but were linked to (1) changes in informant-reported traits and (2) distal outcomes. For example, participants whose personality structures more closely matched a Big Five structure showed socially desirable trait increases (e.g., higher conscientiousness) yet reported lower happiness and life satisfaction a year later. Greater influence of specific indicators (e.g., laziness, depressed mood) predicted better self-rated physical health two years later. These findings complicate assumptions that socially desirable traits are uniformly beneficial and illuminate when and for whom macro-level patterns diverge from micro-level dynamics.

Summary: The Promise and Challenges of Personality Systems

Predicting micro-, meso-, and macro-level outcomes remains a key goal in psychological science. The rise of new technology has meant increasing adoption of prediction research by fields outside psychology. Yet we argue that the multilevel, multivariate systems approach we have laid out highlights the key role that psychology can play in explainable prediction.

Table 2

Broad categories of network parameters and their theoretical implications to outcomes

Network Parameter	Broad Definition	Graphical Depiction	Theoretical Implication
Organization	The degree to which a personality network resembles a certain (theoretical) structure <u>Example Parameter:</u> Normalized Mutual Information		Structures similar to the Big Five may change in more normative ways (e.g., maturity principle) and may be most associated with outcomes commonly associated with the Big Five.
Topology - Network	The number of connections between indicators in the network (e.g., a count) <u>Example Parameter:</u> Density		Denser networks are more sensitive to perturbations and reflect volatile dynamics that are more vulnerable to change (e.g., a shift to a new equilibrium). We would therefore expect them to show more change and be associated with poorer outcomes.
Topology - Indicators (Nodes)	The overall influence of a single indicator in the network <u>Example Parameter:</u> Strength Centrality		The overall influence of a single indicator may have cascading effects on the system as a whole. These indicators may be particularly difficult to change and have outsize impacts on long-term outcomes.
Complexity	The number and magnitude of all the connections between indicators in the network <u>Example Parameter:</u> Entropy		More complex networks are reflective of reactive systems that are also sensitive to very strong changes. Similar to density, we would expect them to show stronger patterns of change and be associated with poorer outcomes.

Note. Indicators (i.e. nodes) are the items inputted into the network and are represented by circles. The lines connecting indicators are called edges, and are the associations between two nodes, accounting for all other associations between nodes. Bolder edges indicate a stronger association between indicators, lighter lines represent weaker associations. Communities are groups of indicators that are strongly associated with each other that other groups of nodes in the network. Communities are represented in the thin ovals and most relevant to network organization.

Conclusion

In this paper, we distinguished three levels of personality prediction, reviewed evidence for each, and outlined opportunities for integrating them bottom-up. We do not view any level as inherently more useful nor fully distinct; rather, each answers different questions. Empirically, it is remarkable that macro-level traits (i.e. patterns across time and contexts) predict outcomes at all, and even more so that they do so consistently. Theoretically, if personality is understood as an emergent hierarchy with causal forces distributed across levels, each level is essential (Jansma & Hoel, 2025). Practically, macro-level traits may identify who is at risk; meso-level states may clarify for whom interventions work; and micro-level dynamics may guide individualized tailoring.

We challenge psychological scientists to consider how integrating heterogeneous micro-level processes with meso- and macro-level constructs could yield more flexible, explainable predictions, illuminating not only who is at risk for long-term outcomes but also what a given person may do and when (see Table 3 for example approaches across different stages of the research process). Such integration may move psychology closer to its goal of describing, predicting, and explaining what people think, feel, and do.

Table 3
Example future directions for integrating personality prediction from the micro- to the macro-level

Research Stage	Example
Assessment	Build traditional psychometric assessments that differentiate individuals based on heterogeneous meso-level patterns and micro-level dynamics (i.e. an example of bottom-up integration at the assessment stage).

Design	Leverage multi-modal and multi-timescale data, such as passive sensing (e.g., actigraphy), environmental data (e.g., pollution), EMA, etc. to capture complex associations across data sources and levels of aggregation.
Modeling	Application of methods used to capture complex associations in dynamic systems, such as differential equations (see e.g., Batra et al., 2025), that capture associations across multiple timescales simultaneously.
Change	Model change in meso-level patterns and micro-level dynamics from measurement burst designs and assess their convergence with macro-level assessments.
Intervention	Compare interventions based on macro-, meso-, and micro-level personality-outcome associations and intervention efficacy.

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[i] Although we use the phrases macro-, meso-, and micro-level to refer to different levels in a personality system, these should not be confuses with Bronfenbrenner's (1979) Ecological Systems Theory, which is concerned with developmental interactions between an individual and their broader context. While our work is compatible with Bronfenbrenner's, ours is largely addressing what he considered the microsystem at different levels of the chronosystem.

[ii] We acknowledge that the boundaries between the meso- and micro-levels can be quite blurry and subject to the question of how much aggregation must take place in order for something to be sufficiently aggregated to be considered meso-level. The nuance of this distinction is complex and unfortunately beyond the scope of a brief paper. But as one example of these complexities, different processes likely unfold at different rates within and across people and processes. Consequently, the "same" process could unfold over minutes for one person and seconds for another. Thus, measuring that process at the level of minutes could be micro-level or meso-level, respectively, for these individuals. So instead, our aim is to provide a useful descriptive framework for thinking about what we are measuring and what kinds of questions this may help us to answer.